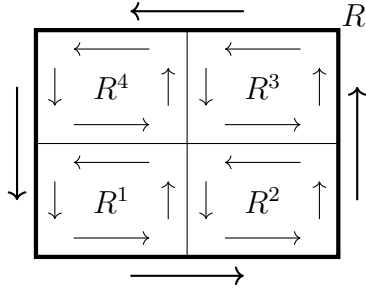


Teo cauchy goursat rectangulos

Teorema 1 (de Cauchy-Goursat para rectángulos).

Sea Ω un dominio en $\mathbb{C}$, $f \in \mathcal{H}(\Omega)$ y $R \subset \Omega$ un rectángulo cerrado $\implies \int_{\partial R} f(z) dz = 0$.

Demostración: Dividimos R en 4 rectángulos $\{R^j\}_{j \in \mathbb{N}_4}$ congruentes (i.e. con el mismo tamaño) y orientamos sus fronteras positivamente como indica el siguiente diagrama:



$$\begin{aligned} \int_{\partial R} f(z) dz &= \sum_{j=1}^4 \int_{\partial R^j} f(z) dz \\ \implies \left| \int_{\partial R} f(z) dz \right| &\leq \sum_{j=1}^4 \left| \int_{\partial R^j} f(z) dz \right|. \end{aligned}$$

Al menos un R^j verifica que $\left| \int_{\partial R^j} f(z) dz \right| \geq \frac{1}{4} \left| \int_{\partial R} f(z) dz \right|$. Lo seleccionamos y lo llamamos R_1 . Repetimos el proceso de división en 4 rectángulos congruentes y orientamos sus fronteras positivamente para seleccionar R_2 . Así sucesivamente, obteniendo una sucesión de rectángulos $R \supset R_1 \supset R_2 \supset \dots \supset R_n \supset \dots$ tal que se cumple que

$$\begin{aligned} \left| \int_{\partial R_n} f(z) dz \right| &\geq \frac{1}{4} \left| \int_{\partial R_{n-1}} f(z) dz \right| \geq \frac{1}{4^n} \left| \int_{\partial R} f(z) dz \right| \\ \implies \left| \int_{\partial R} f(z) dz \right| &\leq 4^n \left| \int_{\partial R_n} f(z) dz \right|. \end{aligned}$$

Veamos que $\left| \int_{\partial R_n} f(z) dz \right| \xrightarrow{n \rightarrow \infty} 0$. Sea $z_0 \in \bigcap_{n=1}^{\infty} R_n \subset \Omega$. Como $f \in \mathcal{H}(\Omega)$, se tiene que $\forall \varepsilon > 0 : \exists \delta > 0 : |z - z_0| < \delta \implies \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon$. Además, $\exists N_0 \in \mathbb{N}$ suficientemente grande tal que $R_{N_0} \subset D(z_0, \delta)$

$$\implies \forall n > N_0 : \forall z \in R_n : |f(z) - f(z_0) - f'(z_0)(z - z_0)| < \varepsilon |z - z_0|.$$

Observamos que $(z)' = 1 \in \mathcal{H}(\Omega) \wedge (z^2/2)' = z \in \mathcal{H}(\Omega) \xrightarrow{\text{RB}} \int_{\partial R_n} 1 \, dz = \int_{\partial R_n} z \, dz = 0$.

$$\begin{aligned} \left| \int_{\partial R_n} f(z) \, dz \right| &= \left| \int_{\partial R_n} (f(z) \pm f(z_0) \pm f'(z_0)(z - z_0)) \, dz \right| \\ &= \left| \int_{\partial R_n} [f(z) - f(z_0) - f'(z_0)(z - z_0)] \, dz + \int_{\partial R_n} [f(z_0) + f'(z_0)(z - z_0)] \, dz \right|. \end{aligned}$$

Luego, por la `prop-abs-integral-linea-compleja-leq-longitud/??`

$$\begin{aligned} \left| \int_{\partial R_n} f(z) \, dz \right| &\leq \int_{\partial R_n} |f(z) - f(z_0) - f'(z_0)(z - z_0)| \, |dz| < \int_{\partial R_n} \varepsilon |z - z_0| \, |dz| \\ &\leq \varepsilon \operatorname{diam}(R_n) \cdot \int_{\partial R_n} |dz| = \varepsilon \operatorname{diam}(R_n) \cdot \operatorname{long}(\partial R_n). \end{aligned}$$

Por lo tanto, $\left| \int_{\partial R} f(z) \, dz \right| \leq 4^n \varepsilon \operatorname{diam}(R_n) \cdot \operatorname{long}(\partial R_n) = \frac{4^n \varepsilon}{2^n} \operatorname{diam}(R) \cdot \frac{1}{2^n} \operatorname{long}(\partial R)$ para $\varepsilon > 0$ arbitrario. Entonces concluimos que $\left| \int_{\partial R} f(z) \, dz \right| = 0$, luego $\int_{\partial R} f(z) \, dz = 0$. ■

Referenciado en

- `teo-cauchy-goursat-rectangulo-singularidades`
- `teo-fn-holomorfa-imp-exists-primitiva`
- `teo-cauchy-goursat-disco`

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.